TMI-Core 0.2: The Axiomatic Core of Automatic Interfaces

The Primary Interface, Definitional Automaticity, and the Transition to Automatically Structured Fields

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Abstract

This document fixes the updated core of the Theory of Manifold Interfaces, denoted by $\mathcal{T}_{MI}^{0.2}$. The central transition of version 0.2 is that automaticity is no longer treated as an external addition to an interface. It is introduced as a definitional property of a valid interface: if a structure cannot autonomously interpret an admissible input and produce an intelligible output for another system, then it is not a full interface, but a passive channel or a component of a meta-interface. The core connects the primary interface axiom, definitional automaticity, the distinction between an interface and a passive channel, derived interfaces, projection into the physical layer, and the scaling of automatically structured interface fields.

Contents

1 Purpose of the Document

The purpose of this document is to fix a minimal core that should not be expanded by adding new major entities. Further work should refine definitions, prove internal consequences, and derive testable physical or engineering projections.

2 Primary Objects

Definition 1 (Null Undifferentiation)**.** *N_0 is the limiting state without distinction, sides, relation, order, or direction. It is not a physical vacuum and not an object of nature, but a limiting symbol of undifferentiation inside the axiomatic scheme.*

Definition 2 (Primary Interface)**.** *I_0 is the first boundary of differentiation through which a minimal distinction arises.*

Definition 3 (First Distinction)**.** *Δ_0 is the minimal result of the action of the primary interface. It unfolds into the minimal two-sided structure $(A, \neg A)$.*

3 The Primary Interface Axiom

Axiom 1 (Primary Interface)**.**

$$\boxed{A_{I_0}: \quad N_0 \xrightarrow{I_0} \Delta_0 \to (A, \neg A).} \tag{1}$$

A shorter form is

$$\boxed{N_0 \xrightarrow{I_0} (A, \neg A).} \tag{2}$$

The meaning of the axiom is that the primary interface does not connect already existing sides; it creates the very possibility of sides. Before I_0, there is no A and no $\neg A$. After the boundary arises, the distinction Δ_0 appears, and this distinction unfolds into the minimal structure $(A, \neg A)$.

4 Valid Interface

Definition 4 (Valid Interface)**.** *A valid interface I is a structure that, in the presence of an admissible input, performs translation, interpretation, or structuring between systems without an external controller directing every act of translation.*

In working form:

$$I_{A\to B} \equiv \big(\partial(A,B), T_I(S_A \to S_B), \mathrm{Interp}_I, \mathrm{Auto}_I\big), \tag{3}$$

where $\partial(A, B)$ is the boundary of interaction, T_I is the state-translation operator, Interp_I is the operation of interpretation, and Auto_I is the self-execution of the interface function under an admissible input.

5 The Axiom of Interface Automaticity

Axiom 2 (Definitional Automaticity)**.**

$$\boxed{A_{\text{auto}}: \quad \forall I \in \mathfrak{I},\ \text{ValidInterface}(I) \Rightarrow \text{Auto}(I).} \tag{4}$$

In short:

$$\boxed{I \in \mathfrak{I} \Rightarrow \text{Auto}(I).} \tag{5}$$

Automaticity does not mean permanent activity:

$$\boxed{\text{Auto}(I) \not\Rightarrow \text{AlwaysAct}(I).} \tag{6}$$

Automaticity means self-activation when an admissible input is present:

$$\boxed{\text{Auto}(I) \equiv \big(\text{Input}(I) \wedge \text{Adm}(I) \wedge \text{Dir}(I) \Rightarrow \text{AutoAct}(I)\big).} \tag{7}$$

6 Passive Channel and Meta-Interface

Definition 5 (Passive Channel)**.** *A passive channel is a structure through which a signal or a state can pass, but which does not perform its own interpretive function without an external controlling agent.*

If translation is controlled by an external agent U_{ext}, then the valid interface is the composite structure

$$\boxed{I' = (I, U_{\text{ext}}).} \tag{8}$$

$$\boxed{\text{PassiveChannel}(I) \wedge \text{Control}(U_{\text{ext}}, I) \Rightarrow \text{Interface}(I').} \tag{9}$$

Theorem 1 (Interface-Channel Distinction)**.** *If a structure does not perform the interface function automatically under an admissible input, then it is either not a valid interface or it is a passive channel included in a broader meta-interface:*

$$\boxed{\neg\,\text{Auto}(I) \Rightarrow \neg\,\text{ValidInterface}(I) \vee \text{PassiveChannel}(I).} \tag{10}$$

Proof. By the axiom of automaticity, a valid interface satisfies $\text{ValidInterface}(I) \Rightarrow \text{Auto}(I)$. Its contrapositive gives $\neg\,\text{Auto}(I) \Rightarrow \neg\,\text{ValidInterface}(I)$. If the structure nevertheless participates in transmission but does not perform translation by itself, then it is a passive channel inside a broader system $I' = (I, U_{\text{ext}})$. Therefore the statement holds inside the axiomatic scheme. □

7 Derived Interfaces

Definition 6 (Derived Interface)**.** $\mathrm{Der}(I, I_0)$ *means that the interface* I *is derived from the primary interface* I_0*, i.e. it belongs to the space of interfaces that becomes possible after the primary differentiation.*

$$\boxed{\mathrm{Der}(I, I_0) \Rightarrow I \in \mathfrak{I} \Rightarrow \mathrm{Auto}(I).} \tag{11}$$

For derived interfaces, automaticity is unfolded conditionally:

$$\boxed{\mathrm{Der}(I, I_0) \Rightarrow \big(\mathrm{Input}(I) \wedge \mathrm{Adm}(I) \wedge \mathrm{Dir}(I) \Rightarrow \mathrm{AutoAct}(I)\big).} \tag{12}$$

8 Bridge to the Physical Layer

If a physical field is an interface field,

$$\Xi_I \in \mathfrak{I}_{\mathrm{phys}}, \tag{13}$$

then it is automatic as an interface:

$$\boxed{\Xi_I \in \mathfrak{I}_{\mathrm{phys}} \Rightarrow \mathrm{Auto}(\Xi_I).} \tag{14}$$

Consequently, the physical interface field should be considered as automatically structured:

$$\boxed{\Xi_I \in \mathfrak{I}_{\mathrm{phys}} \Rightarrow \Xi_I^{\mathrm{auto}}.} \tag{15}$$

9 Projection Chain

The full form of the field is

$$\boxed{\Xi_I^{\mathrm{auto}}.} \tag{16}$$

Its projections are

$$\boxed{\Xi_I^{\mathrm{auto}} \to \Xi_I \to \chi_I.} \tag{17}$$

Here χ_I is the minimal scalar amplitude projection:

$$\boxed{\chi_I = \Pi_{\mathrm{amp}}(\Xi_I^{\mathrm{auto}}).} \tag{18}$$

10 Scaling

A local automatically structured field scales into a network and a manifold:

$$\boxed{\Xi_I^{\mathrm{auto}} \to \{\Xi_{I,k}^{\mathrm{auto}}\}_{k \in K} \to \mathcal{N}_\Xi \to \mathcal{M}_\Xi^{\mathrm{auto}} \to \mathrm{Struct}_I(\mathcal{M}).} \tag{19}$$

11 Unified Core Formula

$$\boxed{N_0 \xrightarrow{I_0} \Delta_0 \to (A, \neg A) \Rightarrow \mathfrak{I} \Rightarrow \mathrm{Auto}(I).} \tag{20}$$

$$\boxed{I_0 \Rightarrow \mathfrak{I} \Rightarrow \mathrm{Auto}(I) \Rightarrow \Xi_I^{\mathrm{auto}} \Rightarrow \mathcal{M}_\Xi^{\mathrm{auto}} \Rightarrow \mathrm{Struct}_I(\mathcal{M}).} \tag{21}$$

12 Status of the Claims

- Level A: axioms and definitions, for example $\mathrm{ValidInterface}(I) \Rightarrow \mathrm{Auto}(I)$.
- Level B: internal theorems, for example the transition from Ξ_I^{auto} to Struct_I under stabilization conditions.
- Level C: physical hypotheses, for example the existence of Ξ_I^{auto} as a physical mechanism.
- Level D: testable consequences, for example an increase of coherence time in a quantum experiment.

13 Conclusion

TMI-Core 0.2 fixes the central spine of the theory: the primary interface creates the possibility of distinction; a valid interface is automatic by definition; physical interface fields are automatically structured; and a collection of such fields forms a scalable interface manifold capable of stabilizing as structure.

References

[1] Salkutsan Aleksey. *Theory of Manifold Interfaces. An Axiomatic Conceptual-Mathematical Framework of Interfaciality, Meaning, and Cognition*. Zenodo. doi:10.5281/zenodo.19940675.

[2] Salkutsan Aleksey. *TMI-Physics 2.2: Automatically Structured Interface Fields*. 2026.